Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Data scientist and quantitative analyst with **8+ years building predictive models and financial data pipelines from first principles**. Built regression models ($R^2 = 0.977$) and **Isolation Forest anomaly detection** at GMU on multi-source datasets; built **Random Forest + XGBoost pipelines on 100K+ records at 85%+ accuracy**. Applied Python ML pipelines to **1.6M financial records** at Entrata with output validation and compliance gates. Expert in Python, SQL, and R; experienced with Tableau and AI workflow automation (Claude/LLM APIs). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Predictive Modeling | Risk Segmentation | Python | SQL | R | XGBoost | Random Forest | Regression Modeling | Anomaly Detection | Tableau | Data Infrastructure | AI Workflow Automation

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Research Associate - Predictive Modeling & Analytics

Fairfax, Virginia

- Built **regression models ($R^2 = 0.977$) from first principles** in Python and R across WHO and UK Road Safety datasets spanning **190+ countries** - multi-source data collection, feature engineering, model validation, and stakeholder reporting.
- Applied **Isolation Forest anomaly detection** to automate integrity checks across 3 research pipelines - surfaced anomalies that manual review missed, **cut manual QA time by 70%**, improved downstream model reliability.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - improved research comprehension across two cohorts by **40%**.

Entrata India (Db Xento Systems)
Oct 2018 - Aug 2024

Software Engineer - Financial Data Infrastructure & Analytics

Pune, India

- Processed **1.6M financial records** through Python and AI transformation pipelines - production data workflows on regulated financial datasets with output validation, quality gates, and compliance checkpoints.
- Designed data infrastructure for 20+ production modules - **200+ SQL queries tuned (30% perf gain)**; PostgreSQL schema design, audit logging, indexing across all client environments.
- Achieved **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines - **zero security incidents**; built 5+ internal analytical tools cutting resolution time by 40%.
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation; containerized 30%+ of microservices with Docker (50% execution time reduction) across 120+ Agile sprints.
- Conducted 300+ code reviews and mentored 5+ engineers on data quality, secure coding, and TDD standards across all major releases.

Entrata India (Db Xento Systems)
Jul 2017 - Sep 2018

Associate Software Engineer

Pune, India

- Improved system stability by **35%** building 20+ full-stack data modules (Python, PostgreSQL) - 99.5% uptime and 25% faster report generation.

PROJECTS

CrashClarity - Risk Factor Modeling & Pattern Detection (104K Records)
2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering | Live: Portfolio

- Achieved **85%+ accuracy** on 104K+ records using Random Forest + Logistic Regression with feature engineering across weather, road, and time risk variables - deployed interactive dashboard for stakeholders.

Real Estate Analysis - Socioeconomic Risk Segmentation (XGBoost)
2025 | Python, XGBoost, Random Forest, Feature Engineering, Pandas

- Analyzed **20+ socioeconomic risk variables** across Connecticut property data (2001-2022) using XGBoost and Random Forest - identified key pricing risk predictors and segmentation drivers.

Global Health & Road Safety - Anomaly Detection & Regression Modeling
2024-2025 | Python, R, Isolation Forest, scikit-learn, Tableau, ggplot2, Quarto | George Mason University

- Built **end-to-end predictive pipelines ($R^2 = 0.977$)** and Isolation Forest anomaly detection across **190+ countries (WHO + UK datasets)** - multi-source ingestion, validation, stakeholder dashboards.

EDUCATION

M.S. Data Analytics Engineering - George Mason University — 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, predictive analytics, ML systems, data infrastructure

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering — 2013-2017

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

ML & Modeling: XGBoost, Random Forest, Logistic Regression, Isolation Forest, scikit-learn, Feature Engineering, Statistical Modeling, PCA, Regression, NumPy, Pandas

Data & Infra: PostgreSQL, MySQL, AWS (S3, EC2, RDS, Glue), Docker, FastAPI, Flask, RabbitMQ; familiar with dbt/Snowflake concepts

Visualization: Tableau, Power BI, ggplot2, Plotly, Quarto, Streamlit (familiar)

AI Tools: LangChain, LangGraph, OpenAI API, Anthropic Claude API, BERT, spaCy, PyTorch, TensorFlow

Process: TDD/BDD, Git, JIRA, Agile/Scrum, secure coding, audit logging, PHI/HIPAA compliance